

**MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING/
SCHOOL OF TECHNOLOGY MANAGEMENT**

Academic Year: 2024-2025

Program/s: BTech/MBATech

Year: II Semester: IV

Stream/s: AI/AI ML/AIDS/CSE(DS)

Subject: Machine Learning

Time: 03 hrs (10:00am to 1:00pm)Date: 28 / 4 / 2025

No. of Pages: 6

Marks: 100

Final Examination/Re-Examination (2022-23, 2023-24)
24-25

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right-hand side indicate full marks.
- 7) Assume Suitable data if necessary.

Q1												
CO-1 ; SO-1 ; BL-2	a.	Compare and contrast Classification and Regression in Supervised Learning. Provide one real-world example for each.	[5]									
CO-1 ; SO-1 ; BL-2	b.	Illustrate the difference between MCAR, MAR, and MNAR in missing data mechanisms. Discuss in detail.	[5]									
CO-2 ; SO-1 ; BL-3	c.	Explain how Regularization techniques improve machine learning models by preventing overfitting. Also, compare Ridge Regression and Lasso Regression techniques of regularization.	[5]									
CO-3 ; SO-1 ; BL-3	d.	Given Confusion matrix <table><tr><td>Actual / Predicted</td><td>Fraud (1)</td><td>No Fraud (0)</td></tr><tr><td>Fraud (1)</td><td>40</td><td>20</td></tr><tr><td>No Fraud (0)</td><td>10</td><td>130</td></tr></table> Compute the precision, recall and F1-score. Interpret the results.	Actual / Predicted	Fraud (1)	No Fraud (0)	Fraud (1)	40	20	No Fraud (0)	10	130	[5]
Actual / Predicted	Fraud (1)	No Fraud (0)										
Fraud (1)	40	20										
No Fraud (0)	10	130										
Q2												
CO-1; SO-1; BL-2	a.	Given the following dataset representing the salaries (in thousands) of employees in a company such as 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 100. 1. Calculate the Interquartile Range (IQR) of the given dataset. 2. Identify any potential outliers using the IQR method.	[5]									

CO-3; SO-1 ; BL-4	b.	How does the bagging ensemble learning approach decrease the variance in decision tree classification?	[5]																						
CO-2; SO-1; BL-3	c.	The article “Relating the Cetane Number of Biodiesel Fuels to Their Fatty Acid Composition: A Critical Study” included the following data on x = iodine value (g) and y = cetane number for a sample of 10 biofuels. The iodine value (x) is the amount of iodine necessary to saturate a sample of 100 g of oil. <table><tr><td>x</td><td>132</td><td>129</td><td>120</td><td>113</td><td>105</td><td>92</td><td>84</td><td>83</td><td>88</td><td>59</td></tr><tr><td>y</td><td>46</td><td>48</td><td>51</td><td>52</td><td>54</td><td>52</td><td>59</td><td>58</td><td>61</td><td>64</td></tr></table> Obtain value of coefficient of determination for the simple linear regression model for best fit line using least square estimators. Also predict number of cetane if iodine value is 125.	x	132	129	120	113	105	92	84	83	88	59	y	46	48	51	52	54	52	59	58	61	64	[10]
x	132	129	120	113	105	92	84	83	88	59															
y	46	48	51	52	54	52	59	58	61	64															
Q3																									
CO-2; SO-1; BL-3	a.	Explain the working of Gradient Descent in machine learning. Given is the dataset for height and weight of children. Using linear regression with gradient descent, compute the value of the Loss (cost) function after two iterations and also determine the new theta values after two iterations. Consider learning rate as 0.001 and initial values of thetas as zero. <table><tr><td>Height (cm)</td><td>Weight (kg)</td></tr><tr><td>140</td><td>35</td></tr><tr><td>150</td><td>40</td></tr><tr><td>160</td><td>50</td></tr><tr><td>170</td><td>60</td></tr><tr><td>180</td><td>70</td></tr></table>	Height (cm)	Weight (kg)	140	35	150	40	160	50	170	60	180	70	[10]										
Height (cm)	Weight (kg)																								
140	35																								
150	40																								
160	50																								
170	60																								
180	70																								
CO-3 ; SO-1 ; BL-4	b.	A study was conducted to classify articles based on the time spent. Article Type is a binary classification where: 0 represents technical article and 1 represents non-technical article. Train a logistic regression model using the given dataset to predict the article type for Time (Hr) = 1.9 for threshold 0.5. Compute confusion matrix and performance metrics (accuracy, precision, recall, F1-score).	[10]																						

			Time (Hr)	Article Type		
			2.7	0		
			1.4	0		
			3.3	1		
			1.3	0		
			3.0	1		
Q4						
CO-1; SO-1; BL-2	a.	Describe the importance of feature selection. Discuss the various feature selection methods to select the most relevant features.				[5]
CO-3; SO-1 ; BL-3	b.	You are given a dataset with two features: Education Level (High School, College) and Internet Usage (High, Low). The target variable is whether a person Buys Online (Yes/No).				[5]
		Record	Education Level	Internet Usage	Buys Online	
		1	High School	High	Yes	
		2	High School	Low	No	
		3	College	High	Yes	
		4	College	Low	No	
		5	College	Low	No	
		Perform the following tasks:				
1. Compute the Gini index of the target variable.						
2. Calculate the Gini index for splitting on the "Education Level" feature.						
3. Calculate the Gini index for splitting on the "Internet Usage" feature.						
4. Which feature should be selected as the root node?						
CO-3; SO-1 ; BL-4	c.	A real estate company wants to classify a new house as Affordable (A) or Expensive (E) based on its size (in square feet) and price using K-NN classification with K = 3.				[10]
		House ID	Size (sq. ft.)	Price (Rs. 100000)	Class	
		A	1500	300	A	
		B	1800	330	A	
		C	2000	355	A	
		D	2200	375	E	
		E	2500	400	E	
		F	2700	420	E	

		A new house has a size of 1900 sq. ft and its price is 340. Using Manhattan Distance, assign the class to the new house. How does the choice of distance metric affect KNN performance?																																													
Q5																																															
CO-3; SO-1 ; BL-3	a.	A retail company wants to segment its customers based on their spending habits. Given the spending scores of six customers, perform two iterations of K-Means clustering (K = 2).	[10]																																												
		<table><tr><th>Customer</th><th>Spending on Product A</th><th>Spending on Product B</th></tr><tr><td>C1</td><td>3</td><td>8</td></tr><tr><td>C2</td><td>4</td><td>7</td></tr><tr><td>C3</td><td>6</td><td>2</td></tr><tr><td>C4</td><td>7</td><td>3</td></tr><tr><td>C5</td><td>5</td><td>6</td></tr><tr><td>C6</td><td>8</td><td>4</td></tr></table>		Customer	Spending on Product A	Spending on Product B	C1	3	8	C2	4	7	C3	6	2	C4	7	3	C5	5	6	C6	8	4																							
		Customer		Spending on Product A	Spending on Product B																																										
		C1		3	8																																										
		C2		4	7																																										
		C3		6	2																																										
		C4		7	3																																										
		C5		5	6																																										
C6	8	4																																													
Initial Centroids:																																															
- Centroid 1 = C1 (3,8) - Centroid 2 = C4 (7,3)																																															
CO-3; SO-1 ; BL-3	b.	A company wants to classify emails as Spam or Not Spam based on the presence of two keywords: "Discount" and "Offer". Given the following dataset:	[10]																																												
		<table><tr><th>Email</th><th>Discount</th><th>Offer</th><th>Spam (1) / Not Spam (0)</th></tr><tr><td>1</td><td>Yes</td><td>No</td><td>Spam (1)</td></tr><tr><td>2</td><td>No</td><td>Yes</td><td>Spam (1)</td></tr><tr><td>3</td><td>No</td><td>Yes</td><td>Not Spam (0)</td></tr><tr><td>4</td><td>Yes</td><td>No</td><td>Not Spam (0)</td></tr><tr><td>5</td><td>No</td><td>Yes</td><td>Not Spam (0)</td></tr><tr><td>6</td><td>Yes</td><td>Yes</td><td>Spam (1)</td></tr><tr><td>7</td><td>No</td><td>No</td><td>Not Spam (0)</td></tr><tr><td>8</td><td>Yes</td><td>No</td><td>Spam (1)</td></tr><tr><td>9</td><td>No</td><td>Yes</td><td>Not Spam (0)</td></tr><tr><td>10</td><td>Yes</td><td>Yes</td><td>Spam (1)</td></tr></table>		Email	Discount	Offer	Spam (1) / Not Spam (0)	1	Yes	No	Spam (1)	2	No	Yes	Spam (1)	3	No	Yes	Not Spam (0)	4	Yes	No	Not Spam (0)	5	No	Yes	Not Spam (0)	6	Yes	Yes	Spam (1)	7	No	No	Not Spam (0)	8	Yes	No	Spam (1)	9	No	Yes	Not Spam (0)	10	Yes	Yes	Spam (1)
		Email		Discount	Offer	Spam (1) / Not Spam (0)																																									
		1		Yes	No	Spam (1)																																									
		2		No	Yes	Spam (1)																																									
		3		No	Yes	Not Spam (0)																																									
		4		Yes	No	Not Spam (0)																																									
		5		No	Yes	Not Spam (0)																																									
		6		Yes	Yes	Spam (1)																																									
		7		No	No	Not Spam (0)																																									
		8		Yes	No	Spam (1)																																									
9	No	Yes	Not Spam (0)																																												
10	Yes	Yes	Spam (1)																																												
Now, a new email contains both "Discount" and "Offer". What is the probability that this email is Spam or Not Spam using the Naïve Bayes classifier?																																															
Q6																																															
CO-3; SO-1 ; BL-3	a.	You are given the following dataset with two features: Weather (Sunny, Overcast, Rainy) and Wind (Weak, Strong). The target variable is whether to Play Tennis (Yes/No).	[10]																																												

		<table><tr><td>Record</td><td>Weather</td><td>Wind</td><td>Play Tennis</td></tr><tr><td>1</td><td>Sunny</td><td>Weak</td><td>No</td></tr><tr><td>2</td><td>Sunny</td><td>Strong</td><td>No</td></tr><tr><td>3</td><td>Overcast</td><td>Weak</td><td>Yes</td></tr><tr><td>4</td><td>Rainy</td><td>Weak</td><td>Yes</td></tr><tr><td>5</td><td>Rainy</td><td>Strong</td><td>No</td></tr><tr><td>6</td><td>Overcast</td><td>Strong</td><td>Yes</td></tr><tr><td>7</td><td>Sunny</td><td>Weak</td><td>No</td></tr><tr><td>8</td><td>Rainy</td><td>Weak</td><td>Yes</td></tr><tr><td>9</td><td>Overcast</td><td>Weak</td><td>Yes</td></tr><tr><td>10</td><td>Sunny</td><td>Strong</td><td>No</td></tr></table> Perform the following tasks: 1. Compute the entropy of the target variable. 2. Calculate the Information Gain (IG) for splitting on the "Weather" feature. 3. Calculate the Information Gain (IG) for splitting on the "Wind" feature. 4. Which feature should be selected as the root node?	Record	Weather	Wind	Play Tennis	1	Sunny	Weak	No	2	Sunny	Strong	No	3	Overcast	Weak	Yes	4	Rainy	Weak	Yes	5	Rainy	Strong	No	6	Overcast	Strong	Yes	7	Sunny	Weak	No	8	Rainy	Weak	Yes	9	Overcast	Weak	Yes	10	Sunny	Strong	No	
Record	Weather	Wind	Play Tennis																																												
1	Sunny	Weak	No																																												
2	Sunny	Strong	No																																												
3	Overcast	Weak	Yes																																												
4	Rainy	Weak	Yes																																												
5	Rainy	Strong	No																																												
6	Overcast	Strong	Yes																																												
7	Sunny	Weak	No																																												
8	Rainy	Weak	Yes																																												
9	Overcast	Weak	Yes																																												
10	Sunny	Strong	No																																												
CO-3; SO-1 ; BL-3	b.	You are given the following dataset: <table><tr><td>Point</td><td>X</td><td>Y</td></tr><tr><td>P1</td><td>2</td><td>2</td></tr><tr><td>P2</td><td>3</td><td>3</td></tr><tr><td>P3</td><td>4</td><td>4</td></tr><tr><td>P4</td><td>10</td><td>10</td></tr><tr><td>P5</td><td>50</td><td>50</td></tr></table> Using DBSCAN with $\epsilon = 2.5$ and MinPts = 2, determine: 1. Core points 2. Border points 3. Outliers 4. Final clusters	Point	X	Y	P1	2	2	P2	3	3	P3	4	4	P4	10	10	P5	50	50	[10]																										
Point	X	Y																																													
P1	2	2																																													
P2	3	3																																													
P3	4	4																																													
P4	10	10																																													
P5	50	50																																													
Q7																																															
CO-1; SO-1; BL-2	a.	Explain the concept of Classification in Supervised Learning. Why is it important? Provide one real-world application.	[5]																																												
CO-3 ; SO-1 ; BL-4	b.	Discuss the key features of the XGBoost ensemble learning approach.	[5]																																												

CO-3; SO-1 ; BL-3	c.	Consider the following distance matrix for six data points (P, Q, R, S, T, U). Apply Agglomerative Clustering with Single Linkage and construct a dendrogram to represent the clustering hierarchy.						[10]
		P	Q	R	S	T	U	
	P	0	3	8	10	7	12	
	Q	3	0	6	9	5	11	
	R	8	6	0	4	2	7	
	S	10	9	4	0	3	6	
	T	7	5	2	3	0	5	
	U	12	11	7	6	5	0	